

DATA ENGINEERING COURSE CURRICULUM

Duration: 3 Months (12 Weeks) | Mode: Instructor-Led + Self-Paced + Projects

Tools Covered: SQL | Python | Cloud Computing (Azure) | Azure Synapse Analytics | Azure Data Factory | Apache Spark | Databricks

MONTH 1: FOUNDATIONS — SQL & DATA MODELING

Week 1: Introduction to Data Engineering & SQL Basics

- What is Data Engineering? Role of a Data Engineer
- Data Lifecycle: ETL vs ELT
- OLTP vs OLAP, Data Warehouse vs Data Lake
- Batch vs Real-Time Processing
- Introduction to RDBMS (MySQL / SQL Server)
- Basic SQL: SELECT, WHERE, ORDER BY, DISTINCT, LIMIT

Week 2: Intermediate SQL

- Aggregate Functions: COUNT, SUM, AVG, MIN, MAX
- GROUP BY and HAVING clauses
- Joins: INNER, LEFT, RIGHT, FULL OUTER
- Subqueries and Nested Queries
- Set Operations: UNION, INTERSECT, EXCEPT

Week 3: Advanced SQL

- Window Functions: RANK, ROW_NUMBER, LEAD, LAG
- Common Table Expressions (CTEs) and Views
- Data Validation & Reconciliation Queries
- Query Optimization Basics
- Hands-on: Business Reporting Queries

Week 4: Data Modeling & Warehousing

- Star Schema vs Snowflake Schema Design
- Slowly Changing Dimensions (SCD Types)
- Fact and Dimension Tables
- Introduction to Data Warehousing Concepts
- Mini Project: Build and Validate a Data Warehouse Schema
- Assessment: SQL & Data Modeling Test

MONTH 2: PYTHON, CLOUD & AZURE SYNAPSE

Week 5: Python for Data Engineering

- Python Basics: Data Types, Loops, Functions
- Working with Pandas & NumPy
- File Handling (CSV, JSON, Parquet)
- API Data Extraction & Automation
- Error Handling, Logging & Config Management
- Mini Project: Python-based ETL Script

Week 6: Cloud Computing Fundamentals (Azure)

- Introduction to Cloud Computing: IaaS, PaaS, SaaS
- Overview of Azure Ecosystem
- Azure Storage: Blob Storage, Data Lake Storage Gen2
- Azure SQL Database Basics
- Azure Portal, Resource Groups & Access Management (IAM)

Week 7: Azure Synapse Analytics

- Introduction to Azure Synapse Analytics
- Synapse Workspace, SQL Pools (Dedicated & Serverless)
- Synapse Pipelines Overview
- Querying Data Lake using Serverless SQL
- Integration of Synapse with Power BI & Spark Pools (Overview)
- Hands-on: Load and Query Data using Synapse SQL Pool

Week 8: Azure Data Factory (ADF) — Fundamentals

- Introduction to ETL/ELT Pipelines in ADF
- ADF Architecture & Components
- Linked Services, Datasets, and Pipelines
- Copy Data Activity: Moving Data Between Sources
- Scheduling and Triggers

MONTH 3: ADF ADVANCED, SPARK, DATABRICKS & CAPSTONE

Week 9: Azure Data Factory — Advanced

- Data Flows: Mapping Data Flows for Transformation
- Integration Runtime Concepts
- Monitoring & Debugging Pipelines
- Connecting ADF with Azure SQL, Synapse, and Blob Storage
- Hands-on: Build an End-to-End Pipeline using ADF

Week 10: Apache Spark & PySpark

- Apache Spark Core Concepts (RDD, DataFrame, Dataset)
- PySpark for Data Transformation
- Spark SQL & Optimization Techniques
- Real-Time Data Processing with Spark Streaming (Overview)
- Project: Transform Big Data using PySpark

Week 11: Databricks

- Introduction to Databricks Workspace & Architecture
- Databricks Notebooks: Clusters, Jobs, and Libraries
- Delta Lake: ACID Transactions, Time Travel, Schema Evolution
- Data Ingestion & Transformation in Databricks
- Integrating Databricks with Azure Data Factory & Synapse
- Hands-on: Build a Data Pipeline using Databricks

Week 12: Capstone Project & Interview Preparation

- Real-Time Capstone Project: SQL + Python + Azure Synapse + ADF + Spark + Databricks Integration (Ingest data → Orchestrate via ADF → Transform with Spark/Databricks → Store & Query in Synapse)
- Cloud Cost Optimization & Pipeline Monitoring
- Resume Building & Portfolio Development (GitHub)
- Mock Interviews & Commonly Asked Interview Questions
- Doubt-Clearing Session & Final Assessment

LEARNING APPROACH

Component	Description
Self-Learning	Pre-reads, recorded sessions, practice exercises
Instructor-Led Live Sessions	Concept walkthroughs and live coding
Hands-on Labs	Weekly assignments and dataset-based practice

Projects	1 project per month + 1 final capstone project
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KEY DELIVERABLES BY COURSE END

- 3 Mini Projects (SQL/Data Modeling, Python ETL, Synapse/ADF Pipeline)
- 1 Capstone Project (End-to-End Cloud Data Engineering Pipeline)
- Resume + Portfolio (GitHub)
- Interview Preparation Kit

TOOLS COVERED

SQL (MySQL/SQL Server) • Python (Pandas, NumPy) • Azure Cloud • Azure Synapse Analytics • Azure Data Factory • Apache Spark (PySpark) • Databricks • Git (optional, for portfolio)